

PROGRAMA DA UNIDADE CURRICULAR	
ANO LETIVO: 2026/27	PROGRAMA: PG em AI & ML

1 – UNIDADE CURRICULAR

UNIDADE CURRICULAR	BLOCO	ÁREA CIENTÍFICA	DURAÇÃO
Time Series Forecasting	2	Inteligência Artificial	13 horas

2 – DOCENTE

NOME	EMAIL
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3 – OBJETIVOS GERAIS

This course provides a comprehensive introduction to time series analysis and forecasting, with a focus on financial, economic, and business applications.

4 – COMPETÊNCIAS A ADQUIRIR

By the end of the course, students will:

- Understand fundamental concepts of time series decomposition, exponential smoothing, and ARMA/ARIMA models.
- Gain hands-on experience in applying forecasting techniques using Python.
- Develop proficiency in advanced methods such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models.
- Critically evaluate and justify methodological choices in real-world forecasting scenarios.

5 – CONTEÚDOS PROGRAMÁTICOS

- Introduction to Time Series Analysis. Time Series Decomposition (Class #0)
- Exponential Smoothing Methods (Class #1)
- ARMA, SARMA, ARIMA and SARIMA Models (Class #2)
- ARIMA Modelling and Forecasting in Practice (Class #3)
- ARIMAX and SARIMAX with Exogenous Variables (Class #4)
- Convolution Neural Networks (CNN), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs) and Gated Recurrent Units (GRUs) (Class # 5)
- Illustrations and Applications using AWS and Python resources (Class #6)

6- MÉTODOS E TÉCNICAS PEDAGÓGICAS

7 – METODOLOGIA DE AVALIAÇÃO

The course assessment is based entirely on a **group project** (3-6 students), which consists of the following components with proposed weightings:

1. Final Project Submission (70%). A comprehensive report (15-20 pages) including:

- Literature review and data description.
- Application of at least two forecasting techniques covered in the course.
- Justification of methodological choices and critical discussion of results.
- Visualizations (tables, graphs) and Python code appendix.

Evaluation criteria: Technical accuracy, originality, clarity, and adherence to project scope.

2. Presentation and Discussion (30%). A 15-minute group presentation followed by a 5-minute Q&A session.

Evaluation criteria: Presentation quality, time management, and ability to defend methodological choices.

CONTÍNUA

MOMENTO	MÉTODO	DATA	PONDERAÇÃO
1			
2			

EXAME NORMAL

MOMENTO	MÉTODO	DATA	PONDERAÇÃO
1			
2			

EXAME RECURSO

MOMENTO	MÉTODO	DATA	PONDERAÇÃO
1			
2			

8- PLANEAMENTO/CRONOGRAMA DAS SESSÕES

DATA/Nº da sessão	CONTEÚDOS	OBJETIVOS PEDAGÓGICOS	MÉTODO PEDAGÓGICO	ATIVIDADES	TEMPO (MINUTOS)
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____/____/____ Nº1					
____/____/____ Nº2					

9 - DOCUMENTAÇÃO DE APOIO

PowerPoint slides, Python scripts, and databases in various formats.

BIBLIOGRAFIA

- Caiado, J. (2022): *Métodos de Previsão em Gestão com Aplicações em Excel*, 3ª Edição, Edições Sílabo, Lisboa.
- Caiado, J. (2024): *Python e Exploração de Dados*, Edições Sílabo, Lisboa.
- Hyndman, R. e Athanasopoulos, G. (2014): *Forecasting: Principles and Practice*, OTexts.com.
- Gloria Gonzalez-Rivera (2013): *Forecasting for Economics and Business*, Pearson.
- Géron, Aurélien (2019): *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd Edition, O'Reilly

NETGRAFIA